IV. Method

A. Method Overview

Chapter III formulates the service recovery problem with user association, beam power, satellite power budget, and EPFD constraints. Under the EPFD constraint, beams from one or more LEO satellites may need to be shut down. A satellite is regarded as critical when at least one of its beams covers the GS region and must be shut down to satisfy the EPFD constraint. It should be noted that the critical satellite is a time-dependent operational role rather than a permanent satellite identity. A satellite is not predefined as critical before entering the near-inline period. It is regarded as critical only during the time slots in which its beams must be shut down due to the GS geometry and the EPFD constraint. After the satellite moves away from the GS direction and its closed beams return to normal operation, it is no longer treated as a critical satellite. Since this criterion is evaluated for all visible LEO satellites, multiple critical satellites may exist in the same time slot. To recover the service of users affected by beam shutdown on critical satellites, this chapter presents an EPFD-aware beam reassociation framework for service recovery.

The proposed framework is designed based on predictable satellite motion, beam footprints, beam-level EPFD contributions, and user service requirements. Instead of assuming a fixed critical-helper geometry, the framework determines the beam roles in each time slot according to the current satellite geometry and EPFD condition. The framework first identifies critical satellites, closed beams, critical safe beams, helper release beams, and helper recovery beams. It then constructs candidate beam relation graphs to describe the possible reassociation relations between critical beams and helper beams.

The proposed framework consists of three major online procedures. First, Safe-Beam Reassociation for Helper Users (SBR) uses the residual capacity of critical safe beams to reassociate part of the helper users originally served by helper release beams. This reduces the loading of helper release beams and releases part of the helper satellite transmit power. Second, the released power is allocated to helper recovery beams according to their priority-weighted recovery demand. Third, Helper-Beam Reassociation for Closed-Beam Users (HBR) uses helper recovery beams to recover closed-beam users. Through these procedures, the proposed framework aims to improve service continuity for closed-beam users while maintaining EPFD compliance and power constraints.

In summary, the proposed EPFD-aware beam reassociation framework consists of the following steps:

- **Time-slot beam role record:** Precompute the critical satellites, closed beams, critical safe beams, helper release beam candidates, helper recovery beam candidates, and candidate beam relation graphs for each time slot.
- **Safe-Beam Reassociation for Helper Users (SBR):** Reassociate selected helper users from helper release beams to critical safe beams, reducing helper beam loading and releasing helper satellite transmit power.
- **Released power allocation for helper recovery beams:** Allocate the released transmit power to helper recovery beams according to the priority-weighted recovery demand.
- **Helper-Beam Reassociation for Closed-Beam Users (HBR):** Reassociate selected closed-beam users to helper recovery beams and recover their service under the available power of helper recovery beams.

B. Time-Slot Beam Role Record

Before the online reassociation procedures are executed, the system constructs a time-slot beam role record for each time slot. This record stores the critical satellites, helper satellites, and the candidate beam relations between critical beams and helper beams in the current time slot. These records are mainly built according to predictable satellite geometry, beam footprints, and beam-level EPFD contributions.

1) Critical Set Identification: For each time slot t, the system first checks whether the aggregate EPFD generated by all active beams of visible LEO satellites satisfies the EPFD limit. If the aggregate EPFD is already below the limit, beam shutdown is not required in this time slot, and the critical satellite set is empty. If the aggregate EPFD exceeds the limit, the system examines the beam-level EPFD contribution of each active beam on all visible LEO satellites and sorts the beams in descending order according to $\mathrm{EPFD}_{s,b}(t)$. The system then sequentially shuts down beams with higher EPFD contributions and updates the aggregate EPFD after each beam shutdown until the aggregate EPFD satisfies the EPFD limit.

Let $\mathcal{B}_s^{\mathrm{off}}(t)$ denote the set of beams shut down on satellite s in time slot t. After the beam shutdown process is completed, all satellites with a non-empty $\mathcal{B}_s^{\mathrm{off}}(t)$ form the critical satellite set, denoted by $\mathcal{S}^{\mathrm{C}}(t)$.

For each critical satellite s, the beams that are not shut down and remain active are denoted by $\mathcal{B}_s^{\mathrm{safe}}(t)$, and are referred to as safe beams. Therefore, the critical satellite set is determined by the beam shutdown result of each time slot. If beams from different satellites must be shut down to satisfy the EPFD constraint, these satellites are included in $\mathcal{S}^{\mathrm{C}}(t)$ at the same time.

2) Helper Set Identification: After critical set identification is completed, the system constructs the helper satellite set, denoted by $\mathcal{S}^{\mathrm{H}}(t)$. Similar to the critical satellite role, the helper satellite is also a time-dependent operational role rather than a permanent satellite identity. A neighboring satellite is regarded as a helper only when its beams have feasible overlap relations with the current critical satellites and can support the service recovery process in the current time slot. When the corresponding critical satellites leave the critical period and their beams return to normal operation, the helper relation is also released. For each helper satellite candidate, the system checks the footprint overlap between its beams and the safe beams and closed beams of critical satellites to determine the possible role of each helper beam.

If the service area of a helper beam overlaps with the footprint of critical safe beams, this helper beam is marked as a helper release beam candidate, denoted by $\mathcal{B}_h^{\mathrm{rls}}(t)$. A helper release beam candidate represents a helper beam whose served users overlap with critical safe beams and whose loading may be reduced through reassociation.

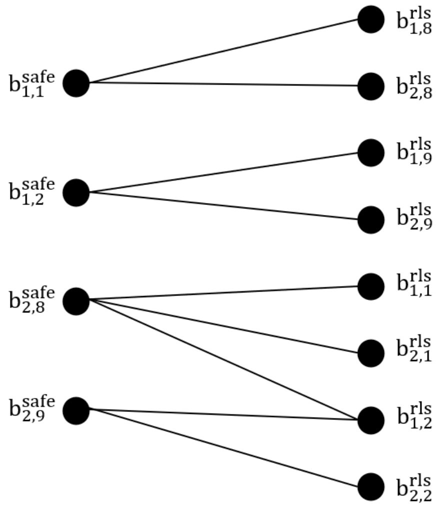

Fig. 1. Safe-release candidate graph between critical safe beams and helper release beam candidates. The left-side nodes $b_{s,\ell}^{\mathrm{safe}}$ denote safe beams on critical satellites, and the right-side nodes $b_{h,k}^{\mathrm{rls}}$ denote helper release beam candidates. An edge $e = (b_{s,\ell}^{\mathrm{safe}}, b_{h,k}^{\mathrm{rls}})$ exists when the two beams have footprint overlap.

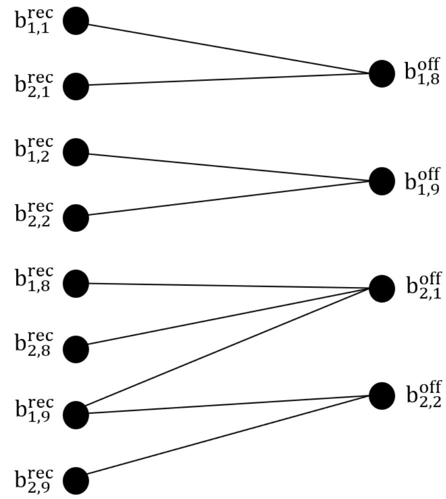

Fig. 2. Closed-recovery candidate graph between critical closed beams and helper recovery beam candidates. The left-side nodes $b_{s,\ell}^{\mathrm{off}}$ denote closed beams on critical satellites, and the right-side nodes $b_{h,k}^{\mathrm{rec}}$ denote helper recovery beam candidates. An edge $e = (b_{s,\ell}^{\mathrm{off}}, b_{h,k}^{\mathrm{rec}})$ exists when the two beams have footprint overlap.

If the service area of a helper beam overlaps with the footprint of critical closed beams, this helper beam is marked as a helper recovery beam candidate, denoted by $\mathcal{B}_h^{\mathrm{rec}}(t)$. A helper recovery beam candidate represents a helper beam whose coverage can include users affected by closed beams.

If a helper satellite h has at least one helper release beam candidate or one helper recovery beam candidate, the satellite is included in the helper satellite set $\mathcal{S}^{\mathrm{H}}(t)$.

3) Candidate Beam Relation Graph Construction: After the critical set and helper set are identified, the system constructs candidate beam relation graphs to record beam-level overlap relations. These graphs indicate which beam pairs have footprint overlap and therefore form candidate reassociation relations.

As shown in Fig. 1, the system first constructs the safe-release candidate graph. One side of this graph consists of critical safe beams, and the other side consists of helper release beam candidates. If the footprint of a critical safe beam overlaps with the service area of a helper release beam, an edge is created between them. Each edge represents a safe-release candidate relation and records the overlap between a critical safe beam and a helper release beam.

As shown in Fig. 2, the system next constructs the closed-recovery candidate graph. One side of this graph consists of critical closed beams, and the other side consists of helper recovery beam candidates. If the footprint of a helper recovery beam overlaps with the service area of a closed beam, an edge is created between them. Each edge represents a closed-recovery candidate relation and records the overlap between a critical closed beam and a helper recovery beam.

Through these graph constructions, the system organizes the possible beam-level relations between critical beams and helper beams before the online reassociation procedures are executed.

It should be noted that the candidate beam relation graphs are used to record feasible reassociation relations, rather than to directly formulate a graph matching problem. An edge in these graphs only indicates that two beams have footprint overlap and may form a candidate reassociation relation. The edge score is not fixed when the graph is constructed. Instead, the actual release score or recovery score is evaluated later according to the current user subset, residual beam capacity, helper beam loading, available helper power, and user satisfaction. After one edge is selected, the user association, remaining capacity, and available helper resources are updated, so the scores of the remaining edges may also change. Therefore, the graph record provides the feasible candidate relations for SBR and HBR, while the final active relations are determined by the following score evaluation and iterative selection procedures.

C. *Safe-Beam Reassociation for Helper Users*

After the time-slot beam role record is completed, the system performs Safe-Beam Reassociation for Helper Users (SBR). The purpose of SBR is to use the residual capacity of critical safe beams to serve part of the users originally served by helper release beams. After these helper users are reassociated to critical safe beams, the loading of the corresponding helper release beams is reduced, allowing the helper satellite to release part of its transmit power.

SBR takes the safe-release candidate graph constructed in the time-slot beam role record as its input. Based on this graph, SBR evaluates the candidate helper users on each safe-release edge and determines the active safe-release relations.

For each edge in the safe-release candidate graph, SBR first identifies the helper users located in the overlap region between the critical safe beam and the helper release beam. Let $\mathcal{U}_e^{\mathrm{H}}(t)$ denote the candidate helper users covered by edge e. These users are originally served by the helper release beam and are located in the overlap region corresponding to the edge. In addition, only helper users that can maintain their service satisfaction after being reassociated to the critical safe beam are included in $\mathcal{U}_e^{\mathrm{H}}(t)$.

Next, SBR calculates how much helper load can be reassociated through the edge according to the residual capacity of the critical safe beam. Let L_u denote the service load of user u, and let $C_e^{\mathrm{safe}}(t)$ denote the residual capacity of the critical safe beam corresponding to edge e after serving its original users. SBR needs to select a feasible subset of helper users from $\mathcal{U}_e^{\mathrm{H}}(t)$, such that the total load of the selected users does not exceed $C_e^{\mathrm{safe}}(t)$ and is as large as possible. Therefore, the release score of edge e is defined as follows:

$$\mathrm{Score}_e^{\mathrm{SBR}}(t) = \max_{\mathcal{A} \subseteq \mathcal{U}_e^{\mathrm{H}}(t)} \sum_{u \in \mathcal{A}} L_u \tag{1}$$

subject to

$$\sum_{u \in \mathcal{A}} L_u \leq C_e^{\mathrm{safe}}(t), \tag{2}$$

where $\mathcal{A}$ denotes a reassociation user subset selected from the candidate helper users. The user subset that achieves this maximum value is denoted by $\mathcal{A}_e^{\mathrm{SBR}}(t)$. Therefore, $\mathrm{Score}_e^{\mathrm{SBR}}(t)$ represents the maximum helper service load that can be reassociated by activating edge e under the current residual capacity of the safe beam. In other words, SBR does not only consider the total helper load on the edge. It further checks which users can actually fit within the residual capacity of the safe beam and how much load can be reassociated.

To select active safe-release relations, SBR performs an iterative greedy activation procedure on the safe-release candidate graph. In each iteration, the system recalculates the release scores of the unselected edges according to the current graph state. The edges that have already been activated are recorded in the active safe-release relation set and are marked as selected in later iterations, so they no longer participate in the next selection. For the currently available edges, the system determines connected components based on whether the edges share critical safe beams or helper release beams. Different connected components do not share the related beam resources and can therefore be processed independently.

For each connected component, SBR selects the edge with the highest release score greater than zero for activation. If multiple edges have the same highest release score, the system randomly selects one of them. For each selected edge, SBR reassociates the helper users in $\mathcal{A}_e^{\mathrm{SBR}}(t)$ to the corresponding critical safe beam. After the reassociation is completed, the edge is recorded as an active safe-release relation.

As shown in Fig. 3, after each iteration, SBR records the edges activated in the current iteration into the active safe-release relation set and updates the original safe-release candidate graph. The updated information includes the user association state, the residual capacity of critical safe beams, the loading of helper release beams, and the released transmit power of helper satellites. The system then recalculates the release scores of the remaining unselected edges according to the updated graph state.

This iterative process continues until the release scores of all unselected edges in the safe-release candidate graph become zero. After SBR is completed, the system obtains the active safe-release relation set, the updated user association state, and the transmit power released by each helper satellite.

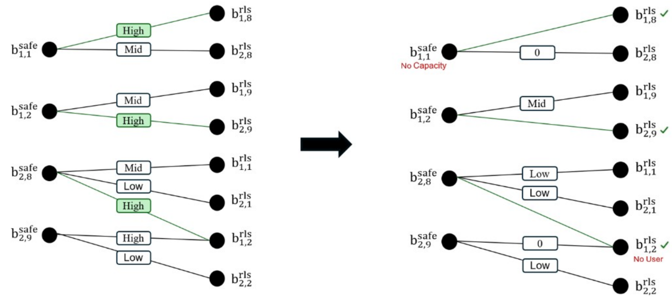

Fig. 3. Example of iterative edge activation and graph update in SBR.

D. Released Power Allocation for Helper Recovery Beams

After SBR is completed, the transmit power released by helper release beams is added back to the available power budget of the corresponding helper satellite. The system then allocates this available power to helper recovery beams on the same helper satellite as the available power of helper recovery beams for serving closed-beam users.

To determine the power allocation order, the system first checks the candidate users connected to each helper recovery beam according to the closed-recovery candidate graph. These candidate users come from critical closed beams that have footprint overlap with the helper recovery beam. The system then calculates the priority-weighted recovery demand of each helper recovery beam based on the service priorities and traffic demands of these candidate users. If a helper recovery beam is connected to more high-priority users or users with larger traffic demand, the beam has a higher recovery demand value.

After the recovery demand value is calculated, the helper recovery beams are sorted in descending order of priority-weighted recovery demand. The released power is then allocated to the helper recovery beams according to this order. Each helper recovery beam can be increased at most to its beam power upper bound. If a beam reaches its upper bound, the remaining released power is allocated to the next helper recovery beam. This process continues until the released power is exhausted or all helper recovery beams have reached their beam power upper bounds.

E. Helper-Beam Reassociation for Closed-Beam Users

After released power allocation is completed, the system performs Helper-Beam Reassociation for Closed-Beam Users (HBR) to recover the service of users located under critical closed beams. The purpose of HBR is to use the available power of helper recovery beams to reassociate users affected by critical beam shutdown to available helper recovery beams.

HBR takes the closed-recovery candidate graph constructed in the time-slot beam role record as its input. Based on this graph, HBR evaluates the candidate closed-beam users on each closed-recovery edge and determines the active closed-recovery relations.

For each edge in the closed-recovery candidate graph, HBR first identifies the closed-beam users located in the overlap region between the critical closed beam and the helper recovery beam. Let $\mathcal{U}_e^{\mathrm{C}}(t)$ denote the candidate closed-beam users covered by edge e. These users are originally located in the service area of a critical closed beam and need to be reassociated to another available beam due to beam shutdown.

Next, HBR determines which closed-beam users can be served by the edge according to the available power of the helper recovery beam. Let $P_e^{\mathrm{avail}}(t)$ denote the available power of the helper recovery beam corresponding to edge e. The users in $\mathcal{U}_e^{\mathrm{C}}(t)$ are first sorted by service priority, where higher-priority users are considered before lower-priority users. Within the same priority level, users with better resource efficiency are considered first, meaning that users requiring less power to achieve the required service satisfaction are served earlier. HBR then sequentially admits users according to this ordering under $P_e^{\mathrm{avail}}(t)$.

Let $\mathcal{A}_e^{\mathrm{HBR}}(t) \subseteq \mathcal{U}_e^{\mathrm{C}}(t)$ denote the set of closed-beam users selected by this priority-ordered sequential admission process. If the remaining power is insufficient to fully satisfy the next ordered user, the user can still be included in $\mathcal{A}_e^{\mathrm{HBR}}(t)$ with partial satisfaction. Therefore, $\mathcal{A}_e^{\mathrm{HBR}}(t)$ contains the users that are reassociated through edge e under the current available power of the helper recovery beam.

Based on $\mathcal{A}_e^{\mathrm{HBR}}(t)$, HBR calculates the recovery score of the edge. Let $w_{\pi(u)}$ denote the weight of the service priority level of user u, and let $s_{u,e}(t)$ denote the satisfaction that user u can achieve after being reassociated through edge e. The recovery score of edge e is defined as follows:

$$\mathrm{Score}_e^{\mathrm{HBR}}(t) = \sum_{u \in \mathcal{A}_e^{\mathrm{HBR}}(t)} w_{\pi(u)} \cdot s_{u,e}(t). \tag{3}$$

This recovery score represents the priority-weighted service recovery benefit that can be achieved by activating edge e under the current available power of the helper recovery beam. If an edge can recover more high-priority users or provide higher weighted satisfaction under the same available power, the edge obtains a higher recovery score.

To select active closed-recovery relations, HBR performs an iterative greedy activation procedure on the closed-recovery candidate graph. In each iteration, the system recalculates the recovery scores of the unselected edges according to the current graph state. The edges that have already been activated are recorded in the active closed-recovery relation set and are marked as selected in later iterations, so they no longer participate in the next selection. For the currently available edges, the system determines connected components based on whether the edges share critical closed beams or helper recovery beams. Different connected components do not share the related beam resources and can therefore be processed independently.

For each connected component, HBR selects the edge with the highest recovery score greater than zero for activation. If multiple edges have the same highest recovery score, the system randomly selects one of them. For each selected edge, HBR reassociates the closed-beam users in $\mathcal{A}_e^{\mathrm{HBR}}(t)$ to the corresponding helper recovery beam. After the reassociation

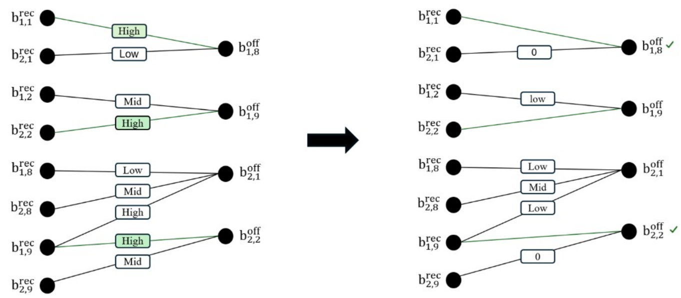

Fig. 4. Example of iterative edge activation and graph update in HBR.

is completed, the edge is recorded as an active closed-recovery relation.

As shown in Fig. 4, after each iteration, HBR records the edges activated in the current iteration into the active closed-recovery relation set and updates the original closed-recovery candidate graph. The updated information includes the user association state, the loading of helper recovery beams, the available power of helper recovery beams, and the closed-beam users that have not yet been recovered. The system then recalculates the recovery scores of the remaining unselected edges according to the updated graph state.

This iterative process continues until the recovery scores of all unselected edges in the closed-recovery candidate graph become zero. After HBR is completed, the system obtains the active closed-recovery relation set, the updated user association state, and the service recovery result of closed-beam users.

F. Overall Procedure and State Update

This section summarizes the overall procedure of EPFD-Aware Beam Reassociation for Service Recovery. The previous subsections describe the time-slot beam role record, SBR, released power allocation, and HBR. This section connects these steps into a complete time-slot procedure and explains the system state that must be updated after each time slot.

Before the online reassociation procedures are executed, the system precomputes the time-slot beam role records for all time slots according to predictable satellite motion, beam footprints, user coverage, and beam-level EPFD contributions. If the aggregate EPFD of a time slot already satisfies the EPFD limit, beam shutdown is not required in that time slot. If the aggregate EPFD exceeds the limit, the corresponding critical satellites, closed beams, critical safe beams, helper release beam candidates, and helper recovery beam candidates are stored in the record.

After the beam role records are completed, the system further constructs the safe-release candidate graphs and the closed-recovery candidate graphs. The safe-release candidate graph is used as the input of SBR to determine which helper users may be reassociated to critical safe beams, thereby releasing helper satellite transmit power. The closed-recovery candidate graph is used as the input of HBR to determine which closed-beam users may be reassociated to helper recovery beams.

In each time slot, the system retrieves the corresponding time-slot beam role record. If no closed beam exists in the time slot, the system does not need to execute the following reassociation procedures. If closed beams exist, the system first performs SBR. SBR selects active safe-release relations according to the release score of each safe-release edge and reassociates selected helper users to the corresponding critical safe beams. After SBR is completed, the loading of helper release beams is reduced, and the corresponding released transmit power is added to the available power budget of the helper satellite.

Before HBR is executed, the system performs released power allocation. This step allocates the available power of the helper satellite to helper recovery beams according to their priority-weighted recovery demand. After the allocation, each helper recovery beam obtains an updated available power.

Finally, the system performs HBR. HBR selects active closed-recovery relations according to the recovery score of each closed-recovery edge and reassociates selected closed-beam users to the corresponding helper recovery beams.

Algorithm 1 Overall Procedure of EPFD-Aware Beam Reassociation for Service Recovery

Input: LEO constellation and beam configuration, GSO–GS and EPFD model, user distribution with service demand and priority weights, and beam power configuration
Output: User association sequence $\mathbf{X}$ and beam power allocation sequence $\mathbf{P}$

Initialize $\mathbf{X}(t)$ and $\mathbf{P}(t)$
Precompute the time-slot beam role records for all time slots
Construct the safe-release candidate graphs for all time slots
Construct the closed-recovery candidate graphs for all time slots
for each time slot t **do**
  Retrieve the time-slot beam role record at time slot t
  if no closed beam exists **then**
    continue
  end if
  // Safe-Beam Reassociation for Helper Users
  Initialize the active safe-release relation set as empty
  Evaluate the release score of each safe-release edge by (1)
  while there exists a safe-release edge with a positive release score **do**
    Divide the current safe-release candidate graph into connected components
    for each connected component **do**
      Select the edge with the maximum release score
      Reassociate the helper users in $\mathcal{A}_e^{\mathrm{SBR}}(t)$ to the corresponding critical safe beam
      Add this edge to the active safe-release relation set
      Mark this edge as selected
    end for
    Update the safe-release candidate graph and recompute the release scores by (1)
  end while
  // Released Power Allocation for Helper Recovery Beams
  Compute the priority-weighted recovery demand of each helper recovery beam
  Sort helper recovery beams in descending order of recovery demand
  Allocate the released power to helper recovery beams subject to beam and satellite power limits
  Update the available power of helper recovery beams
  // Helper-Beam Reassociation for Closed-Beam Users
  Initialize the active closed-recovery relation set as empty
  Evaluate the recovery score of each closed-recovery edge by (3)
  while there exists a closed-recovery edge with a positive recovery score **do**
    Divide the current closed-recovery candidate graph into connected components
    for each connected component **do**
      Select the edge with the maximum recovery score
      Reassociate the closed-beam users in $\mathcal{A}_e^{\mathrm{HBR}}(t)$ to the corresponding helper recovery beam
      Add this edge to the active closed-recovery relation set
      Mark this edge as selected
    end for
    Update the closed-recovery candidate graph and recompute the recovery scores by (3)
  end while
  Update $\mathbf{X}(t)$ and $\mathbf{P}(t)$
end for
return $\mathbf{X}$, $\mathbf{P}$

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